Cybersecurity - Graduate Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. ____________________________ began to show up few years back on wireless access points as a new way of adding or connecting new devices.
   - (a) WPA 2
   - (b) WPA
   - (c) WPS
   - (d) WEP
2. How do you prevent SQL injection?
   - (a) Escape queries
   - (b) Interrupt requests
   - (c) Merge tables
   - (d) All of the above
3. Which among them has the strongest wireless security?
   - (a) WEP
   - (b) WPA
   - (c) WPA 2
   - (d) WPA 3
4. How does a buffer overflow on the stack facilitate running attacker-injected code?
   - (a) By overwriting the return address to point to the location of that code
   - (b) By writing directly to the instruction pointer register the address of the code
   - (c) By writing directly to %eax the address of the code
   - (d) By changing the name of the running executable, stored on the stack
5. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
   - (a) Joined
   - (b) Authenticated
   - (c) Submitted
   - (d) Shared

True / False (10 marks)

Answer True or False

6. True or False: Brute-force attacks on access credentials occur primarily at the session layer of the OSI model.
7. True or False: When the STP solver times out on a constraint query, EXE assumes the query is not satisfiable and halts execution of the path.
8. True or False: Unauthorized network access is not classified as a vulnerability specific to the transport layer.
9. True or False: WPA primarily relies on AES encryption for securing wireless networks instead of TKIP.
10. True or False: A 5-bit MAC can still provide security if the underlying cryptographic algorithm is strong enough.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the vulnerabilities of SQL queries to injection attacks, analyzing how user input can be exploited and the implications for cybersecurity practices.
12. Discuss the significance of various block cipher operating modes in cybersecurity, and critically analyze why certain options, such as CBF, do not qualify as valid modes.

End of Paper